

Problem EV Market Report

1. Introduction

Project Details

- **Project Name:** EV Market Analysis
- **Group Name:** AKASH GROUP
- **Team Leader:** AKASH BAURI

Electric Vehicles (EVs) are transforming the automotive industry, offering sustainable alternatives to traditional fuel-powered vehicles. As the world moves toward cleaner energy solutions, the EV market has experienced rapid growth, driven by technological advancements, government policies, and environmental concerns.

This project, "**EV Market Analysis**," aims to analyze trends, challenges, and opportunities in the EV industry using data-driven insights, machine learning, and visualization techniques. By examining adoption rates, sales trends, registration data, and market patterns, we seek to identify key factors influencing EV growth and predict future industry trends. Our goal is to provide valuable insights that can help stakeholders, policymakers, and businesses make informed decisions in the evolving EV landscape.

2. Dataset Overview

This project utilizes multiple datasets to analyze key trends in the EV market, ensuring accuracy through data cleaning and preprocessing. The datasets include:

- **EV Manufacturers Dataset:** Contains details about EV makers, their locations, and the states where they operate.
- **Total Registration Dataset:** Provides data on the number of EVs registered across different vehicle classes.
- **Daily Registrations Dataset:** Tracks daily registration trends of EVs to understand market fluctuations.
- **EV Market Dataset:** Includes crucial information such as vehicle models, manufacturers, battery capacity, range per charge, pricing details, and sales data over time.

Using **Python, SQL, and Power BI**, the data was analyzed to uncover market trends, adoption patterns, and consumer preferences. The primary goal is to extract meaningful insights, identify key factors influencing EV growth, and predict future market trends.

3. Project Tasks & Methodology

Data Cleaning & Preprocessing

- Renamed columns for better readability.
- Removed unnecessary spaces and characters.
- Converted numerical values stored as strings to integers.
- Handled missing values and inconsistencies.
- Standardized numerical and categorical data using LabelEncoder.

Exploratory Data Analysis (EDA)

- Analyzed sales trends, pricing patterns, and battery performance.
- Identified key factors influencing EV adoption.

Data Visualization & Insights

- Used **Power BI** and **Python libraries** like Matplotlib and Seaborn to generate visualizations.
- Visualized key metrics such as **sales trends, manufacturer-wise distribution, and battery efficiency.**

Time Series Analysis

- Applied **seasonal decomposition** to analyze trends in EV registrations.

Machine Learning Model Implementation

- Used **RandomForestRegressor** to predict future EV registration trends.
- Built a **predictive model** to estimate future EV sales.
- Split the dataset into **training and testing sets** to evaluate model performance using accuracy metrics.

These tasks provided valuable insights into market trends, consumer behavior, and future growth projections in the EV industry.

4. Data Visualization & Insights

Data visualization played a crucial role in understanding trends, patterns, and key insights in the EV market. Various methods and tools were used to make the findings more comprehensible and actionable. The key visualization techniques include:

- **Seaborn & Matplotlib:** Used to create various charts and graphs for data analysis.
- **Line Charts:** Analyzed EV sales growth and registration trends over time.
- **Bar Charts:** Compared manufacturer-wise market shares and regional EV adoption.
- **Scatter Plots:** Examined correlations between battery capacity, range, and pricing.
- **Heatmaps:** Highlighted regional adoption trends and market concentration.
- **Trend Analysis:** Visualized the increase in EV registrations and sales patterns over time.

- **Correlation Analysis:** Identified relationships between various factors affecting the EV market.

These visualizations provided clear insights, revealed hidden patterns, and helped in making informed predictions about the EV industry's future growth.

5. Machine Learning Model Overview

To predict EV sales and registration trends, a **Regression Model** was implemented, specifically using the **RandomForestRegressor** due to its efficiency and accuracy. This model was selected for the following reasons:

- **Handles Large Datasets Efficiently:** Processes vast amounts of data without significant performance loss.
- **Accurate Predictions:** Combines multiple decision trees to improve forecast reliability.
- **Works Well with Categorical & Numerical Data:** Effectively captures relationships between key factors such as price, battery efficiency, and demand.
- **Sales & Registration Forecasting:** Helps policymakers and manufacturers make informed decisions based on future projections.

The model was evaluated using **Mean Squared Error (MSE)** and **R-squared value** to ensure reliability and accuracy. The insights gained from this predictive analysis provide a data-driven approach to understanding EV market trends, assisting stakeholders in strategic planning and decision-making.

Conclusion

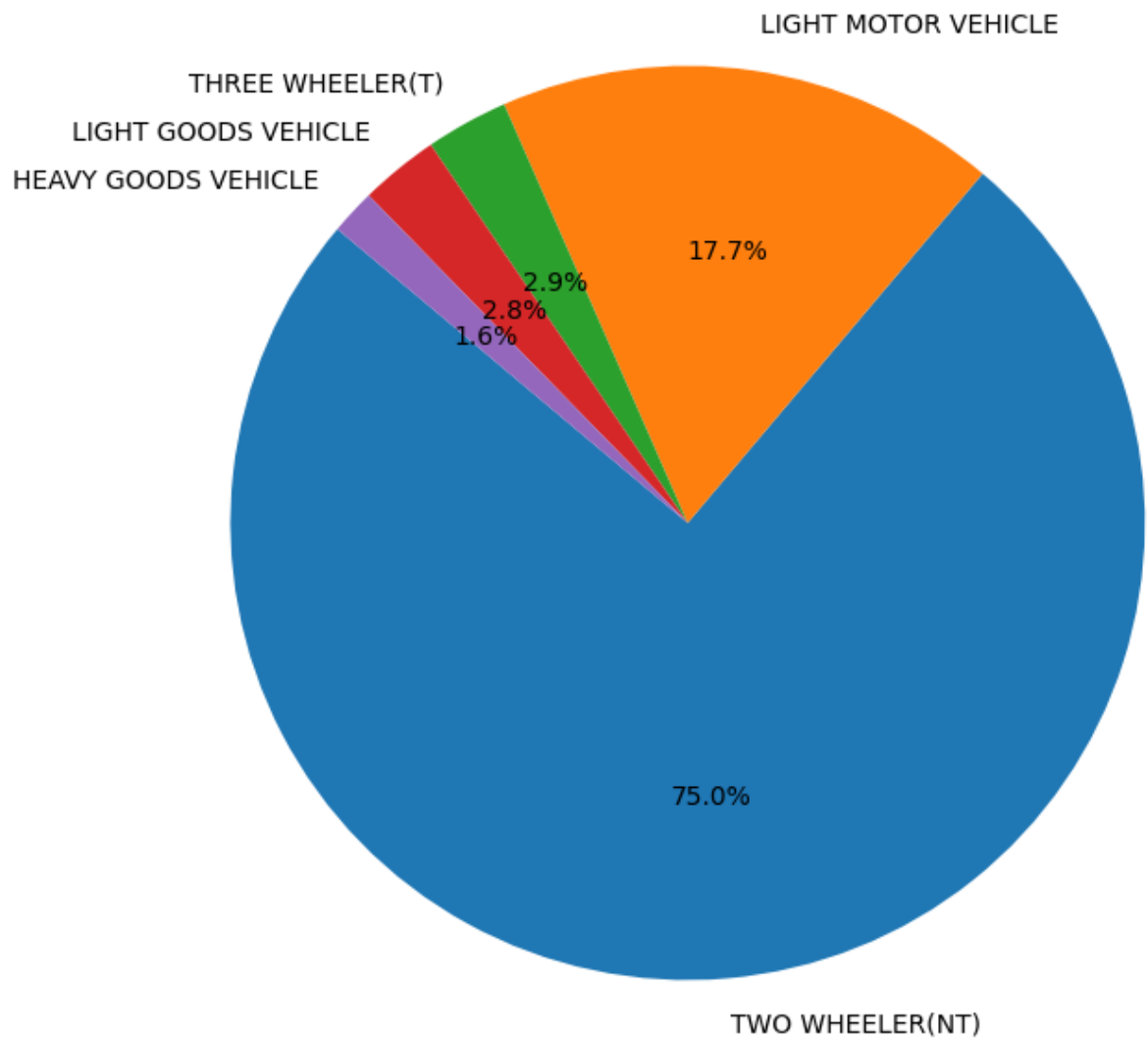
This project provided valuable insights into EV market trends, consumer behavior, and future growth projections using data analysis, visualization, and machine learning techniques. The findings will assist policymakers, manufacturers, and businesses in making data-driven decisions to support the continued expansion of the EV industry.

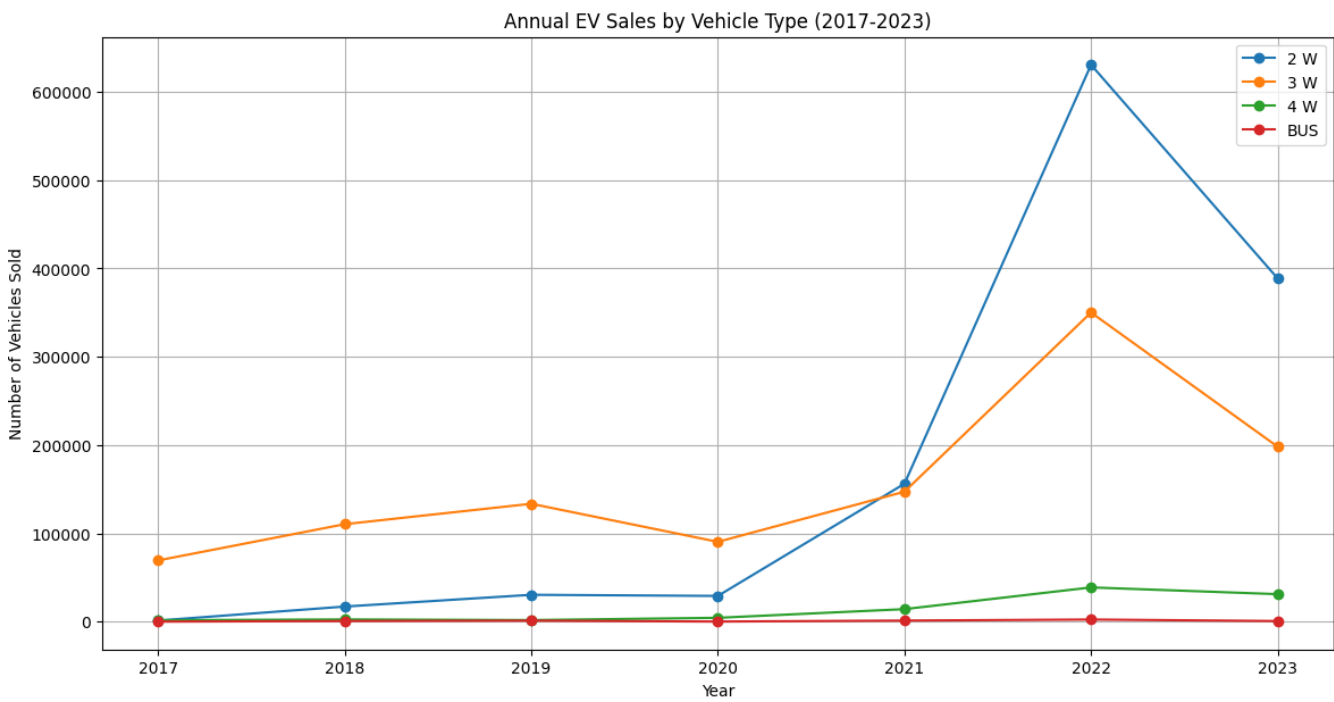
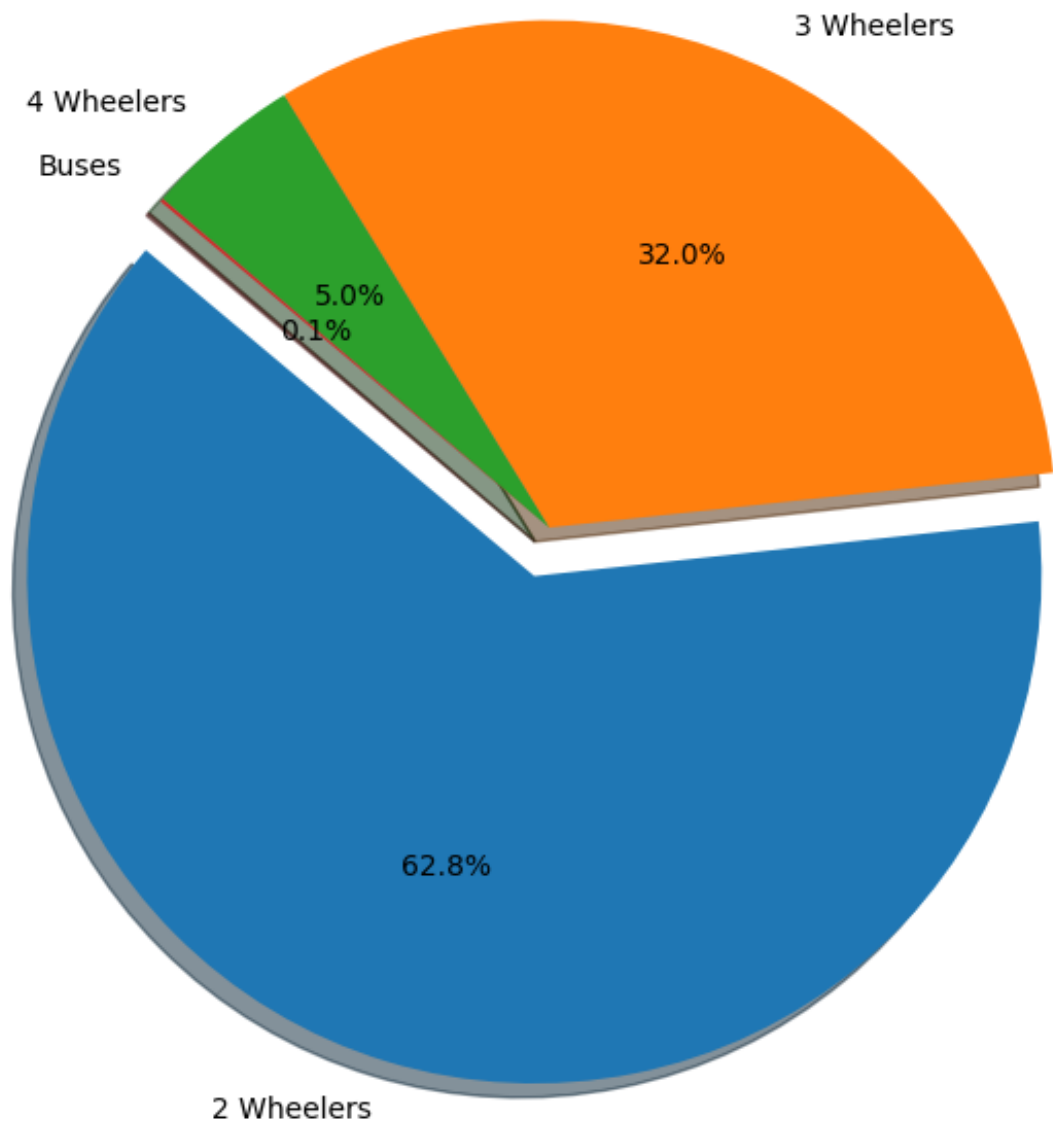
Dataset Links:

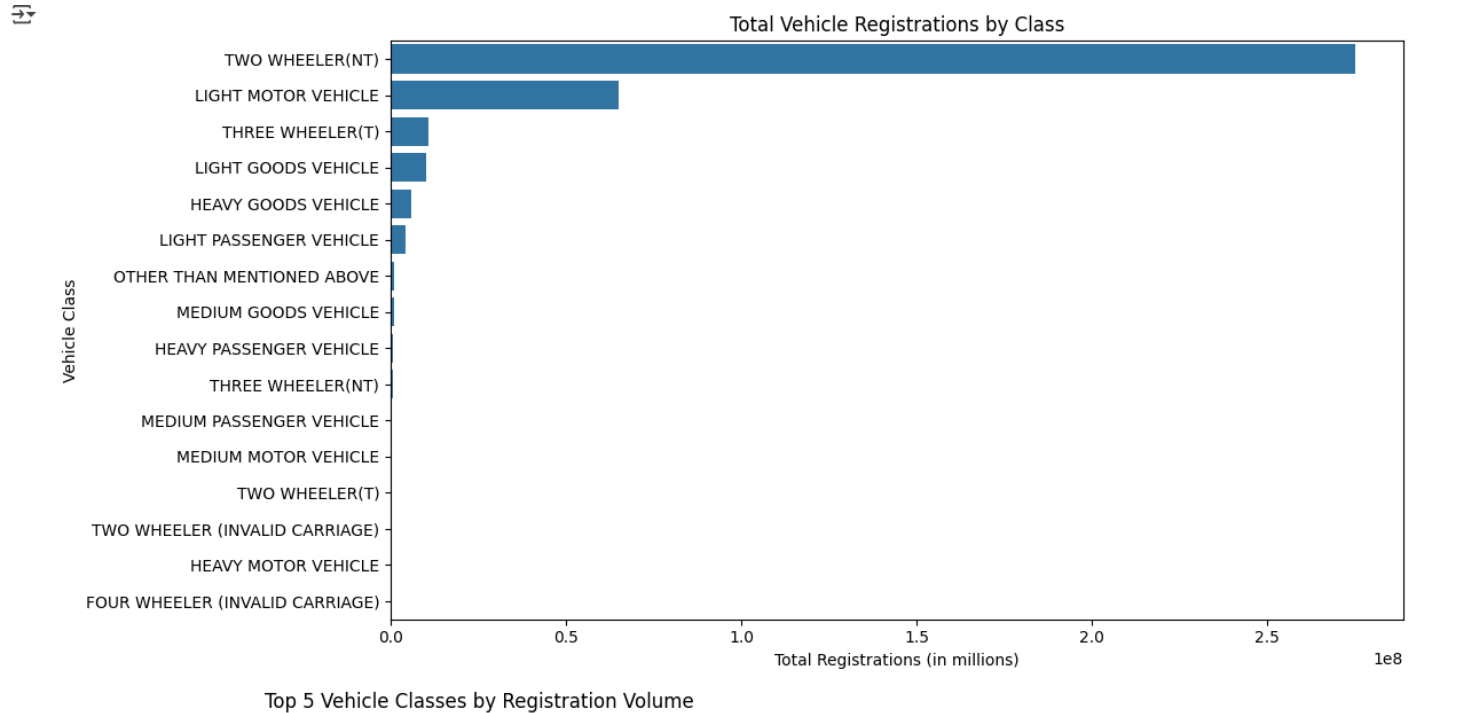
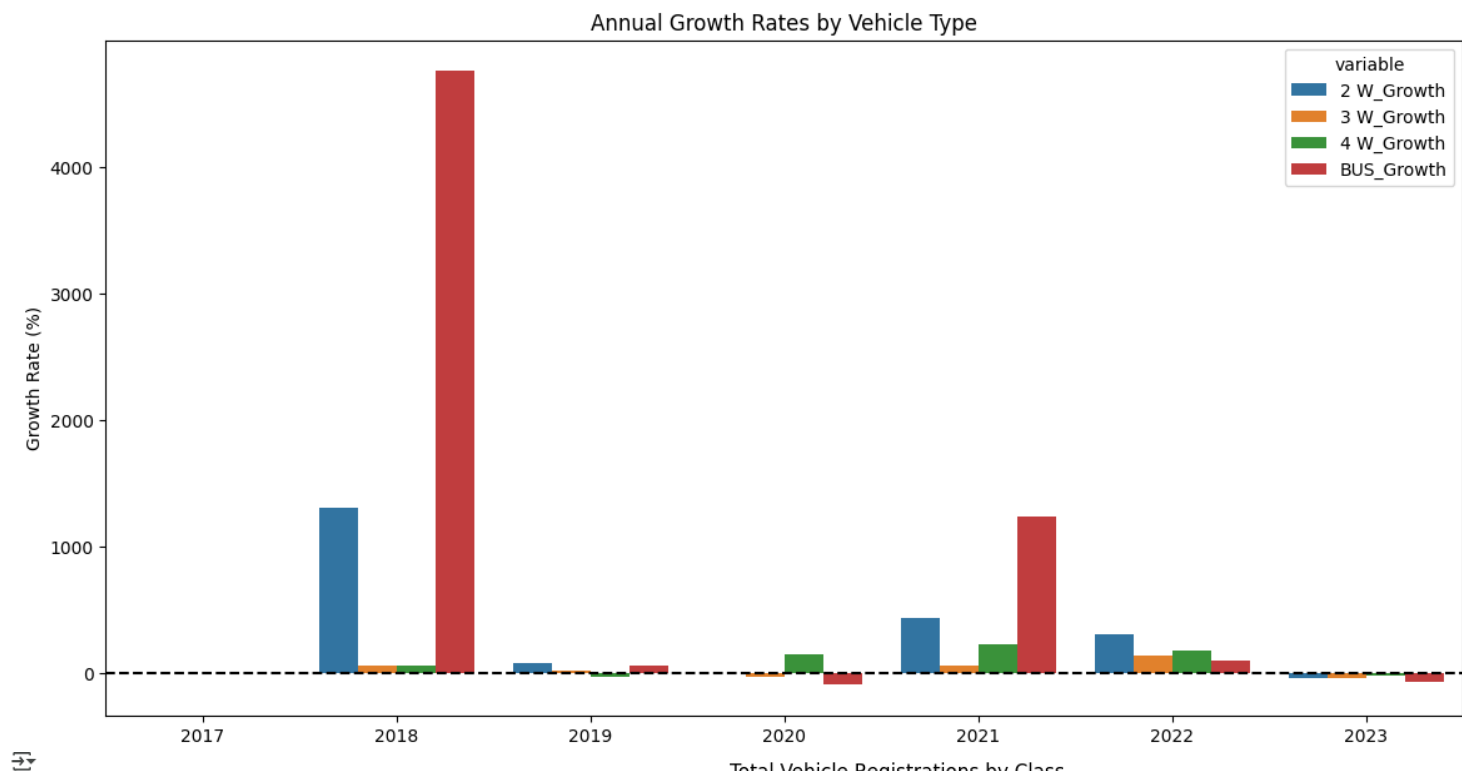
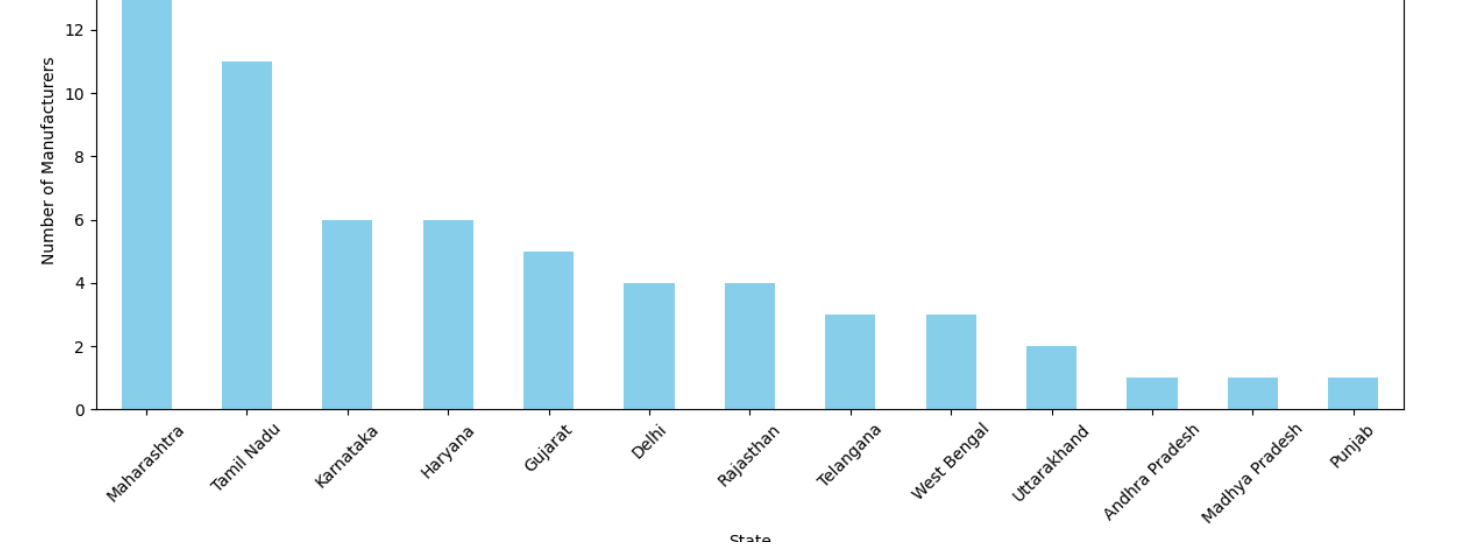
1. https://colab.research.google.com/drive/1WFDk3V9E02QwXf4g1GcSp_pAep592T-a?usp=sha
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Top 5 Vehicle Classes by Registration Volume





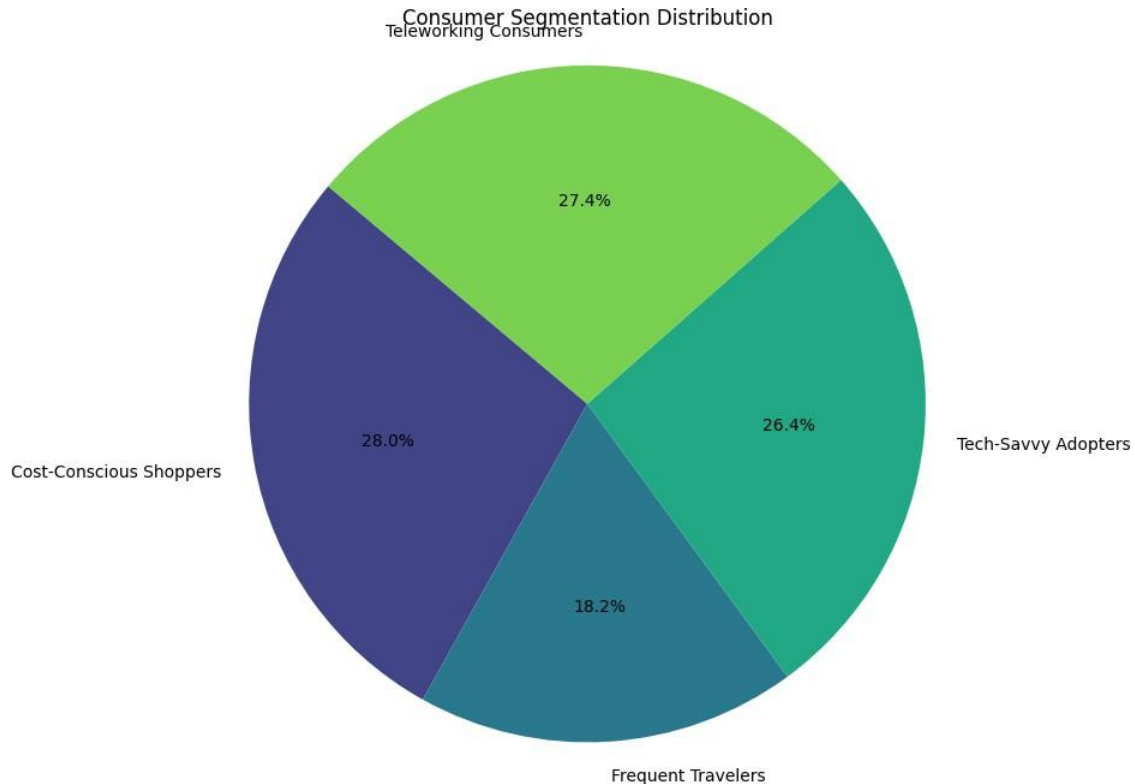


EV Market Behavioral Analysis

Introduction

Consumer behavior analysis reveals patterns in travel, shopping, and digital engagement. This report presents key findings in a visually intuitive format for decision-making.

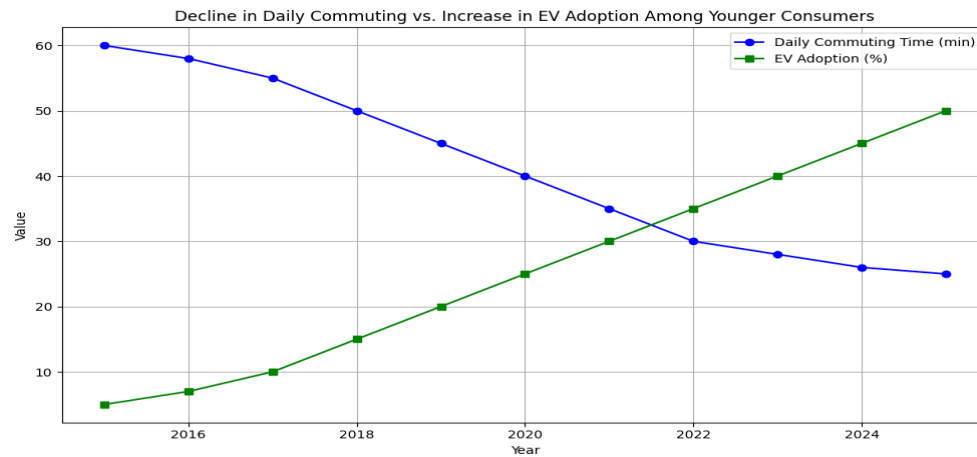
1.1 Consumer Segmentation



Key consumer groups based on behavior:

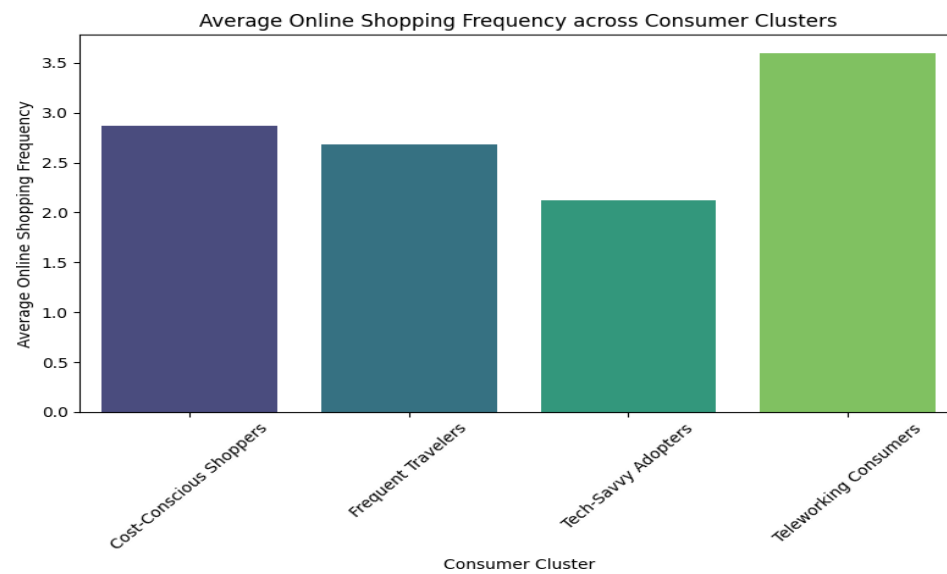
- Frequent Travelers – Prioritize efficiency and convenience.
- Remote Workers – Increased reliance on digital services and online shopping.
- Cost-Conscious Shoppers – Actively seek discounts and compare options.
- Tech-Savvy Adopters – Engage more with digital platforms and electric vehicles.

1.2 Travel and Transportation Trends



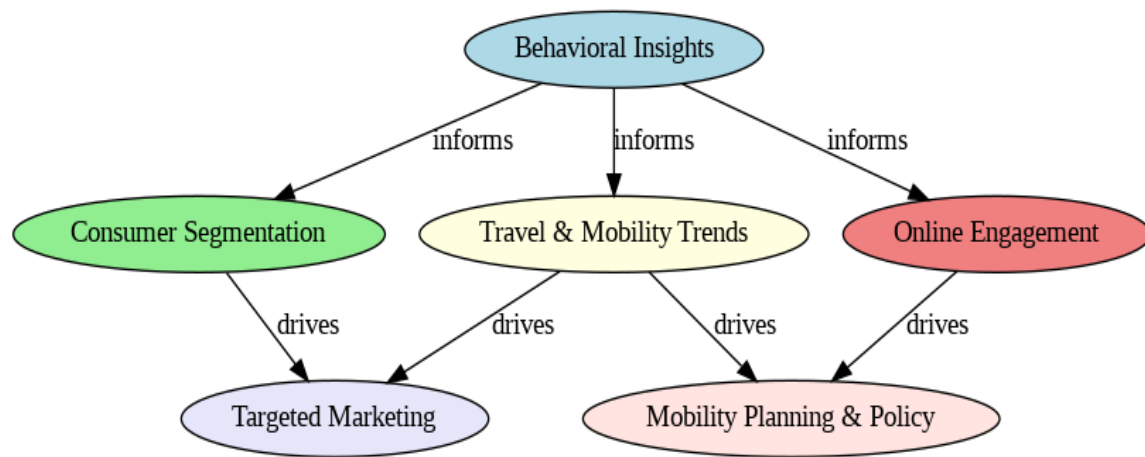
- Decline in daily commuting due to remote work.
- Growth in electric vehicle adoption among younger consumers.
- Increased reliance on ride-sharing and sustainable mobility solutions

1.3 Online Shopping & Digital Engagement



- Teleworkers shop online more frequently.
- Tech-savvy consumers prefer app-based transactions.
- Discounts and promotions attract cost-conscious buyers.

1.4 Business & Policy Implications



- Targeted marketing – Personalized promotions based on consumer segments.
- Smart mobility planning – Improve urban infrastructure for evolving commuting trends.
- AI-driven recommendations – Enhance online shopping experience.
- Sustainability focus – Promote electric vehicles and eco-friendly practices.

Conclusion

Behavioral analysis highlights evolving consumer habits. Businesses and policymakers can use these insights for tailored marketing, better urban planning, and improved service delivery.

Github_link: <https://github.com/adi302/feynn-labs>

Aakash Bauri Team
Market Segmentation Report
By Suraj Mahajan

1. Introduction

The Indian electric vehicle (EV) market is undergoing a transformative phase, driven by a combination of environmental awareness, government initiatives, and technological advancements. With increasing fuel prices and stringent emission norms, consumers are gradually shifting towards sustainable mobility solutions. This report provides an in-depth demographic segmentation analysis of the EV market in India, focusing on geographic distribution, income levels, urban vs. rural adoption, and consumer behavior patterns.

Objectives of the Report

- identify key demographic segments influencing EV adoption.
- Analyze regional manufacturing hubs and demand centers.
- Assess price sensitivity and affordability across income groups.
- Evaluate the impact of government policies on market growth.
- Provide actionable recommendations for automakers, policymakers, and investors.

Scope of the Report

This report covers:

- Two-wheelers and four-wheelers (including luxury EVs).
- State-wise manufacturing and demand trends.
- Consumer preferences based on age, income, and lifestyle.
- Competitive analysis of major EV brands.

2. Methodology

Data Sources

- EV_list.xlsx: Contains specifications, pricing, and consumer ratings of popular 2-wheeler and 4-wheeler EVs in India.
- EV_CARS_INDIA.csv: Provides technical details (battery capacity, range, acceleration) of select 4-wheeler EVs.
- EV Maker by Place.csv: Lists EV manufacturers and their operational locations across India.

Analytical Approach

1. Quantitative Analysis:
 - Price segmentation (budget, mid-range, premium, luxury).
 - Geographic clustering of manufacturing units.
 - Demand estimation based on infrastructure and policy support.
2. Qualitative Analysis:
 - Consumer behavior trends (urban vs. rural).
 - Brand perception and loyalty.
 - Impact of subsidies and incentives.

Limitations

- Limited data on commercial EVs (e.g., e-buses, e-trucks).
- No granular city-level demand breakdown.

3. Demographic Segmentation Analysis

3.1 Geographic Segmentation (State-wise Distribution)

Manufacturing Hubs

State	Key Manufactures	Market Shares (%)
Maharashtra	Tata Motors, Bajaj Auto, Audi, Mercedes	28%
Tamil Nadu	Ola Electric, Hyundai, TVS, Ashok Leyland	22%
Karnataka	Ather Energy, Mahindra Electric	18%
Gujarat	MG Motor, Triton Electric	12%
Haryana	Revolt Motors, Maruti Suzuki	10%

Key Insights:

- Maharashtra and Tamil Nadu dominate due to industrial infrastructure.
- Karnataka is a tech-driven hub for startups like Ather and Simple Energy.
- Gujarat is emerging with MG Motor and new battery plant

Demand Hotspot

- Metro Cities (Delhi, Mumbai, Bengaluru): High adoption due to charging stations and subsidies.
- Southern States (Karnataka, Tamil Nadu): Strong policy push (e.g., Bengaluru’s EV-only delivery zones).
- Emerging Markets (Rajasthan, Uttar Pradesh): Growth in low-speed electric scooters.

3.2 Urban vs. Rural Market Potential

Urban Market Characteristics

- Preferred Vehicles: High-speed EVs (e.g., Ola S1 Pro, Tata Nexon EV).
- Charging Infrastructure: 70% of India's charging stations are in urban areas.
- Consumer Profile: Tech-savvy, environmentally conscious professionals.

Rural Market Characteristics:

- Preferred Vehicles: Low-cost 2-wheelers (e.g., Kinetic E-Luna, Ampere Reo).
- Barriers: Limited charging infra, range anxiety.
- Opportunities: Solar-powered charging, battery-swapping models.

Comparative Snapshot:

Metric	Urban	Rural
EV Penetration	65%	15%
Growth Rate (CAGR)	₹3–10 lakh	₹50,000–1.5 lakh
Growth Rate (CAGR)	25%	12%

3.3 Income-Based Segmentation (Price Sensitivity)

Four Key Segments

1. Budget (Under ₹1 Lakh):

- Buyers: Students, gig workers.
- Examples: Lectrix SX25, Yo Edge.

2. Mid-Range (₹1–3 Lakh):

- Buyers: Young professionals.
- Examples: Ola S1 X, TVS iQube.

3. Premium (₹3–10 Lakh):

- Buyers: Upper-middle-class families.
- Examples: Tata Nexon EV, MG ZS EV.

4. Luxury (₹10+ Lakh):

- Buyers: Affluent individuals, corporates.
- Examples: Audi e-tron GT, Mercedes EQS.

Price Sensitivity Analysis:

- 70% of buyers opt for mid-range EVs.
- Luxury EVs contribute <5% to sales but drive brand innovation.

3.4 Age and Lifestyle Segmentation

Age Group	Preferences	Top Models
18–30	Stylish, high-speed 2-wheelers	Ola S1 Pro, Ather 450X
30–45	Family-friendly 4-wheelers	Tata Nexon EV, MG Comet
45+	Comfort, brand trust	Hyundai Kona, Mercedes EQC

Behavioral Insights:

- Youth (18–30): Prioritize tech features (app connectivity, fast charging).
- Families (30–45): Focus on safety (airbags, range reliability).
- Senior Buyers (45+): Prefer established brands (Hyundai, Tata).

4. Psychographic & Behavioral Insights

- Eco-Conscious Buyers: 40% choose EVs for environmental benefits.
- Cost-Savers: 35% prioritize fuel savings over upfront cost.
- Tech Enthusiasts: 25% adopt EVs for smart features (OTA updates).

5. Government Policies & Incentives

- FAME II Scheme: Subsidies up to ₹1.5 lakh for EVs.
- State-Level Policies:
 - Delhi: Waiver on road tax, registration fees.
 - Karnataka: Subsidies for EV manufacturing.

6. Competitive Landscape

Brand	Market Share	Key Strength
Tata Motors	32%	Affordable 4-wheelers
Ola Electric	25%	High-speed 2-wheelers
Ather Energy	15%	Premium scooters

7. Key Findings

1. Maharashtra and Tamil Nadu are EV manufacturing leaders.
2. Urban consumers drive 70% of sales.
3. Mid-range EVs (₹1–3L) dominate the market.
4. Youth (18–30) are the fastest-growing segment.

8. Strategic Recommendations

1. Expand Rural Reach: Partner with local dealers for affordable EVs.
2. Enhance Charging Infrastructure: Solar-powered stations in Tier 2/3 cities.
3. Target Gig Economy: Customized EVs for delivery fleets.

9. Conclusion

India's EV market is highly segmented, with urban, youth, and mid-income buyers leading adoption. Strategic focus on affordability, infrastructure, and policy alignment will accelerate growth.

10. Appendix

Data Tables: State-wise manufacturing, price segmentation.

Github: <https://github.com/itssun12/EV-market-Segmentation-.git>

Aakash Bauri Team

Market Segmentation Report (geographical dataset)

By Sreehari . s

Introduction

The electric vehicle (EV) market in India is experiencing rapid growth due to increasing environmental concerns, government incentives, and technological advancements. This report presents a clustering analysis of the Indian EV market using machine learning techniques to identify key market segments.

Data Processing and Cleaning

The dataset used in this analysis was preprocessed by handling missing values, encoding categorical variables, and standardizing numerical features. Missing numerical values were replaced with their median, while categorical values were filled with their mode. Label Encoding was applied to categorical features, and numerical data was standardized using StandardScaler to ensure equal weightage for clustering.

Exploratory Data Analysis (EDA)

Feature Correlation Analysis

A heatmap of feature correlations was generated to examine relationships between different attributes. Strong correlations among key factors can help identify significant market trends.

Visualization of Data Distribution

- Pairplots were used to examine relationships between numerical features.
- Histograms displayed the distribution of numerical features.
- Boxplots helped detect outliers in the dataset.

Clustering Analysis

K-Means Clustering

To segment the EV market, K-Means clustering was applied. The optimal number of clusters (K) was determined using the Elbow Method, which analyzes the sum of squared errors (SSE) across different values of K. Based on the elbow plot, the optimal number of clusters was determined to be **4**.

Silhouette Score Evaluation

A silhouette score was calculated to evaluate the quality of the clustering. The silhouette score measures how well each data point fits within its assigned cluster, with a higher score indicating better-defined clusters.

Principal Component Analysis (PCA) for Visualization

To visualize the clustered data, PCA was applied to reduce the dataset to two dimensions. The resulting PCA scatter plot provides insights into the segmentation of the EV market.

Key Findings and Market Segmentation

Based on clustering results, the Indian EV market can be broadly segmented into four clusters:

1. **High-End EV Segment:** Includes premium electric cars with advanced technology, high battery range, and premium pricing.
2. **Mid-Range EV Segment:** Covers vehicles with moderate pricing, decent battery capacity, and government subsidies making them accessible to middle-income groups.
3. **Two-Wheeler and Small EV Segment:** Encompasses affordable electric scooters and three-wheelers catering to urban mobility and last-mile connectivity.
4. **Commercial and Fleet Vehicles:** Includes electric buses, trucks, and fleet vehicles used for public transport and logistics.

Conclusion and Recommendations

The clustering analysis reveals distinct segments in India's EV market, highlighting opportunities for businesses and policymakers. The findings suggest that:

- The premium segment is growing, but affordability remains a key concern for mass adoption.

- Government incentives play a crucial role in expanding the mid-range segment.
- The two-wheeler EV segment is witnessing rapid adoption due to rising fuel prices and environmental awareness.
- Investment in EV infrastructure, including charging stations, will further accelerate market growth.

The insights from this study can help stakeholders make data-driven decisions to enhance market strategies and foster sustainable EV adoption in India.

Clustering analysis completed successfully.

GITHUB : https://github.com/sreeharis1912/EV_India